

Exercise: Nested For Loop

1. Right-Angle Triangle Number Pattern

Write a Java program to display the pattern like a right angle triangle with a number.

Test Data

Input number of rows : 10

Expected Output :

```
1
12
123
1234
12345
123456
1234567
12345678
123456789
12345678910
```

2. Right-Angle Triangle Repeating Numbers

Write a program in Java to make such a pattern like a right angle triangle with a number which repeats a number in a row.

The pattern is as follows :

```
1
22
333
4444
```

3. Right-Angle Triangle with Incrementing Numbers

Write a Java program to make such a pattern like a right angle triangle with the number increased by 1.

The pattern like :

```
1
2 3
4 5 6
7 8 9 10
```

4. Pyramid with Repeating Numbers

Write a Java program to make such a pattern like a pyramid with a number that repeats in the same row.

```
1
2 2
3 3 3
4 4 4 4
```

5. Print Floyd's Triangle

Write a Java program to print Floyd's Triangle.

Test Data

Input number of rows : 5

Expected Output :

```
Input number of rows : 5
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

6. Display Diamond Pattern

Write a Java program to display the pattern like a diamond.

Test Data

Input number of rows (half of the diamond) : 7

Expected Output :

```

  *
 * *
* * *
* * * *
* * * * *
* * * * * *
* * * * * * *
* * * * * * * *
* * * * * * * *
* * * * * * *
* * * * * *
* * * * *
* * * *
* * *
* *
*
```

7. Reverse * Triangle

Write a Java program to generate the following * triangles.

Test Data

Input the number: 6

Expected Output :

```
* * * * *
* * * *
* * *
* *
*

```

8. Right-Angle Triangle with @ Symbols

Write a Java program to generate the following @'s triangle.

Test Data

Input the number: 6

Expected Output :

```
@
@@
@@@
@@@@
@@@@@
@@@@@
@@@@@
```